ELE101

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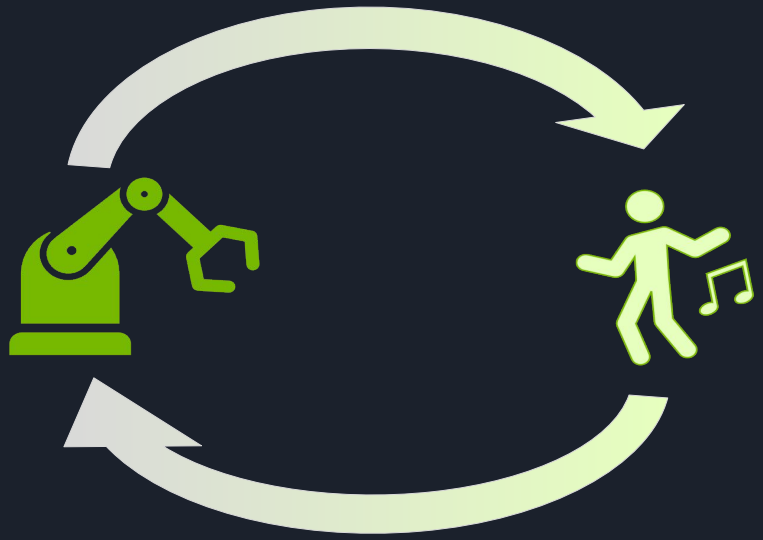
Outline

- AI Introduction
- NVIDIA DLI
- ELE495

Human Vs Machine Learning

Relaxed Alertness



Human	Machine
Rest and Digest	Training
Fight-or-flight	Prediction

Beginning of Artificial Intelligence

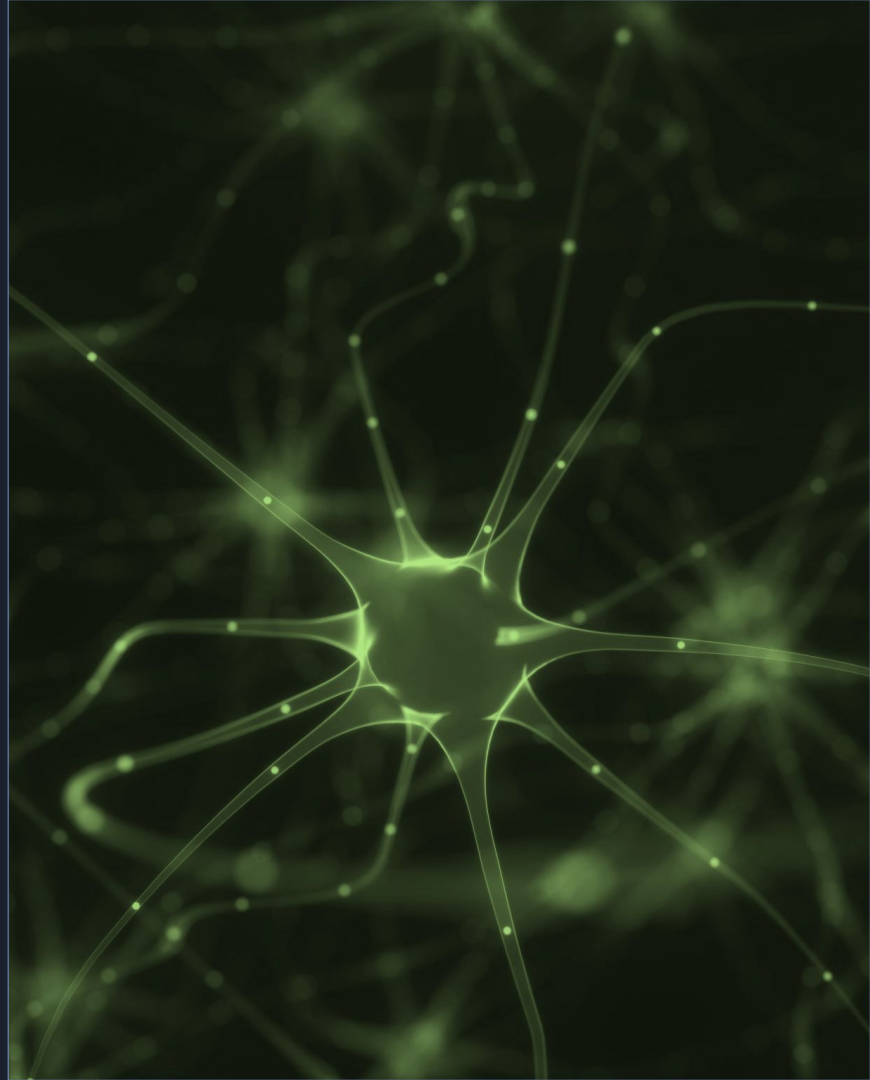
Computers are made in part to complete human tasks

Early on, generalized intelligence looked possible

Turned out to be harder than expected

Early Neural Networks

- Inspired by biology
- Created in the 1950’s
- Outclassed by Von Neumann Architecture

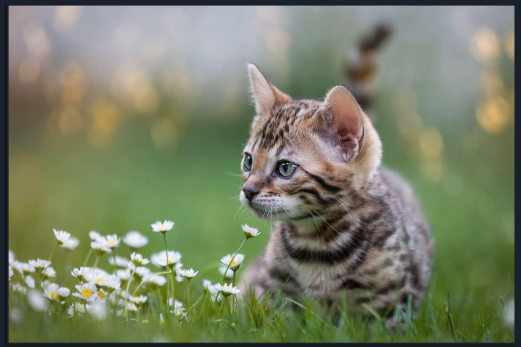


Expert Systems

- Highly complex
- Programmed by hundreds of engineers
- Rigorous programming of many rules

Expert Systems - Limitations

What are these three images?



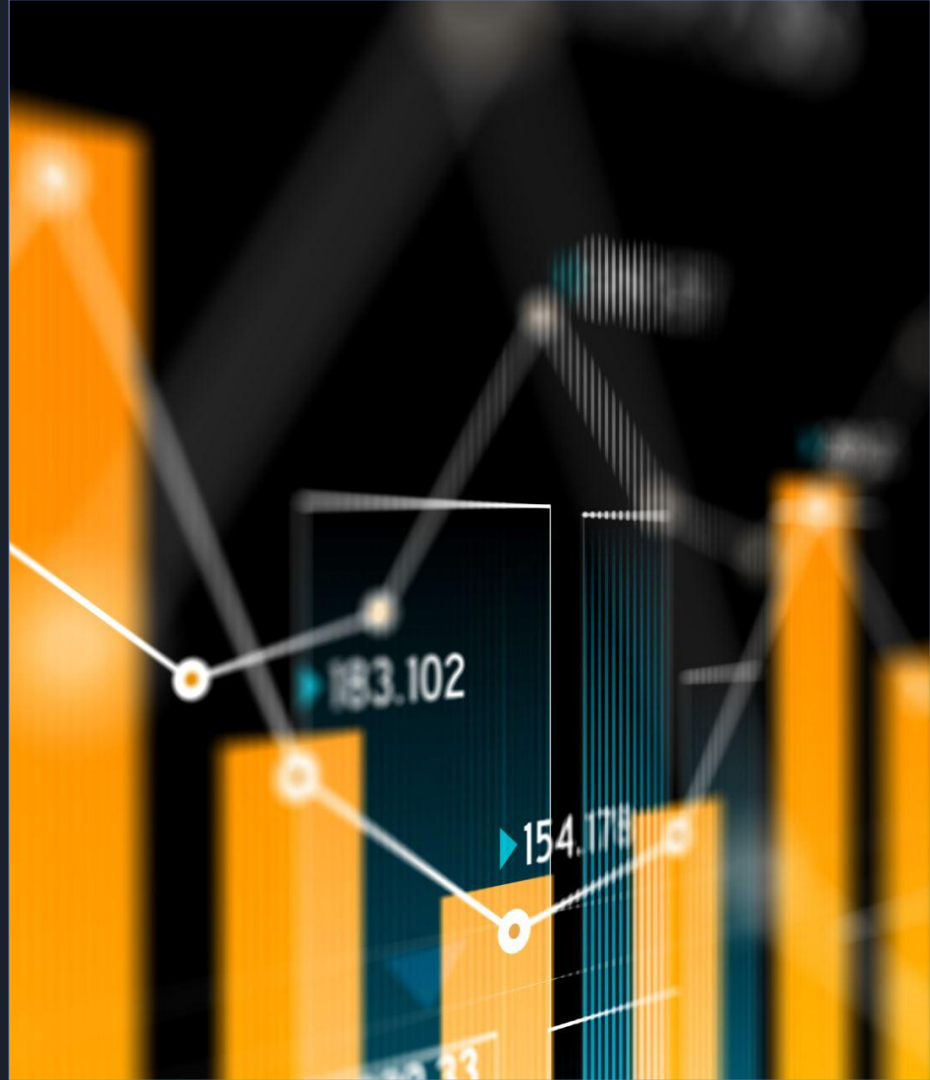
How Do Children Learn?

- Expose them to lots of data
- Give them the “correct answer”
- They will pick up the important patterns on their own

The Deep Learning Revolution

Data

- Networks need a lot of information to learn from
- The digital era and the internet has supplied that data



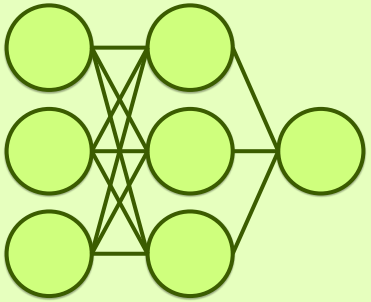
Computing Power

Need a way for our artificial “brain” to observe lots of data within a practical amount of time.

The Importance of the GPU

A Rendered Image

A Neural Network



What is Deep Learning?

“

Deep learning flips traditional programming on its head

”

Traditional Programming

Building a Classifier

1. Define a set of rules for classification
2. Program those rules into the computer
3. Feed it examples, and the program uses the rules to classify

Machine Learning

Building a Classifier

1. Show model the examples with the answer of how to classify
2. Model takes guesses, we tell it if it’s right or not
3. Model learns to correctly categorize as it’s training. The system learns the rules on its own

This is a Fundamental Shift

When to Choose Deep Learning

Classic Programming

If rules are clear and straightforward, often better to program it

Deep Learning

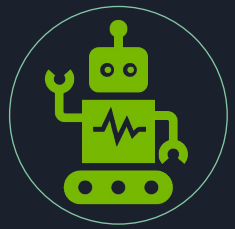
If rules are nuanced, complex, difficult to discern, use deep learning

Deep Learning Compared to Other AI

- Depth and complexity of networks
- Up to billions of parameters (and growing)
- Many layers in a model
- Important for learning complex rules

How Deep Learning is Transforming the World

Computer Vision



ROBOTICS AND MANUFACTURIN G

OBJECT DETECTION

SELF-DRIVING CARS

Natural Language Processing

REAL-TIME
TRANSLATION

VOICE
RECOGNITION

VIRTUAL
ASSISTANTS

Recommender Systems

CONTENT
CURATION

TARGETED
ADVERTISING

SHOPPING

Reinforcement Learning

ALPHAGO BEATS WORLD CHAMPION IN GO

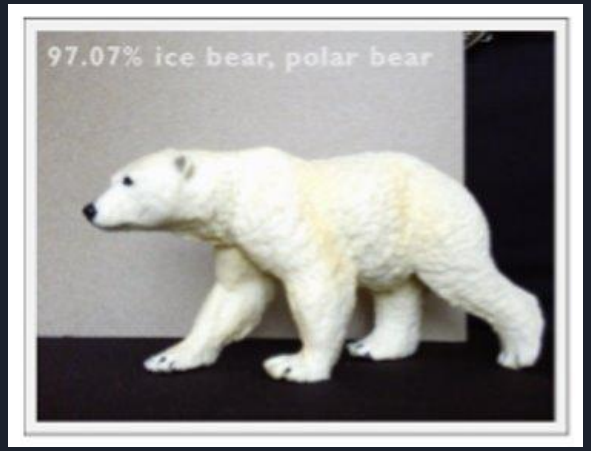
AI BOTS BEAT PROFESSIONAL VIDEOGAMERS

STOCK TRADING ROBOTS

Image Classification

Objective

- Recognize the main subject in the image and assign a label to the image



Datasets for Image Classification

Architectures

- LeNet
- AlexNet
- VGG-16
- Inception V1
- ResNet-50
- Xception
- MobileNet

Object Detection

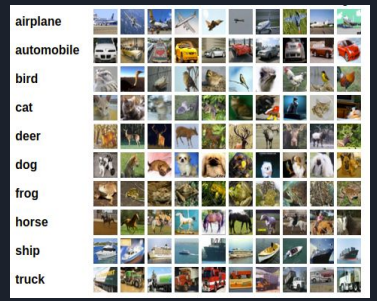
Objective

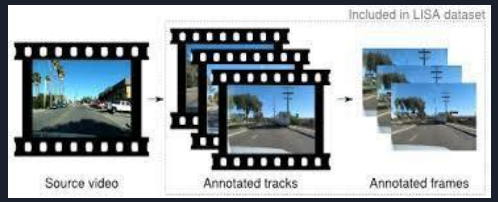
- Predict the class and the position of multiple objects in the image

Datasets for object detection

Architectures

- Sliding Window
- R-CNN
- Fast R-CNN
- Faster R-CNN
- YOLO
- SSD
- Transformers





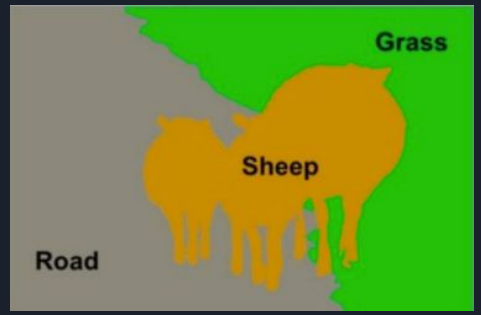
Semantic Segmentation

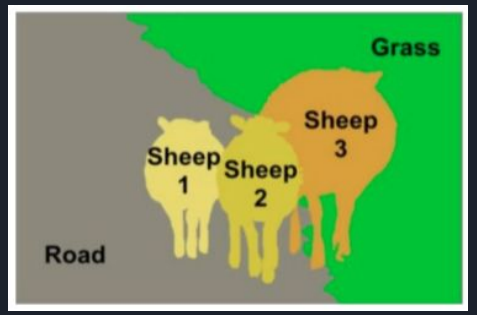
Semantic Segmentation Objective:

- Classify all pixels
- Don't differentiate instances.

Instance Segmentation Objective:

- Classify pixels but differentiate instances (predict segmentation mask)

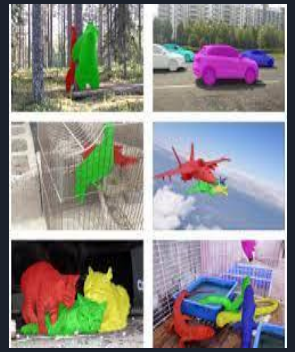




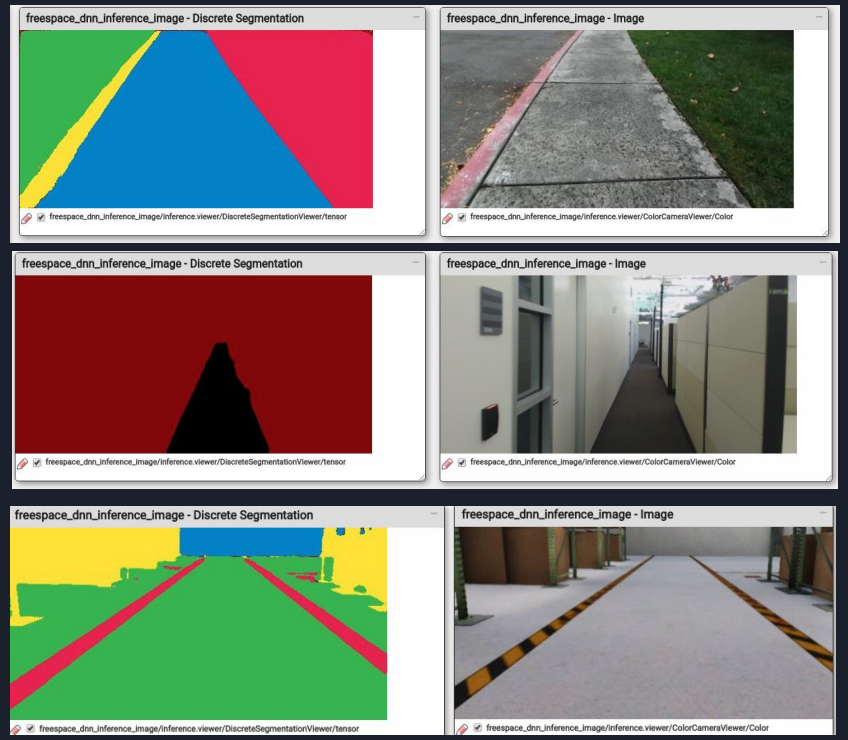
Datasets for Semantic Segmentation

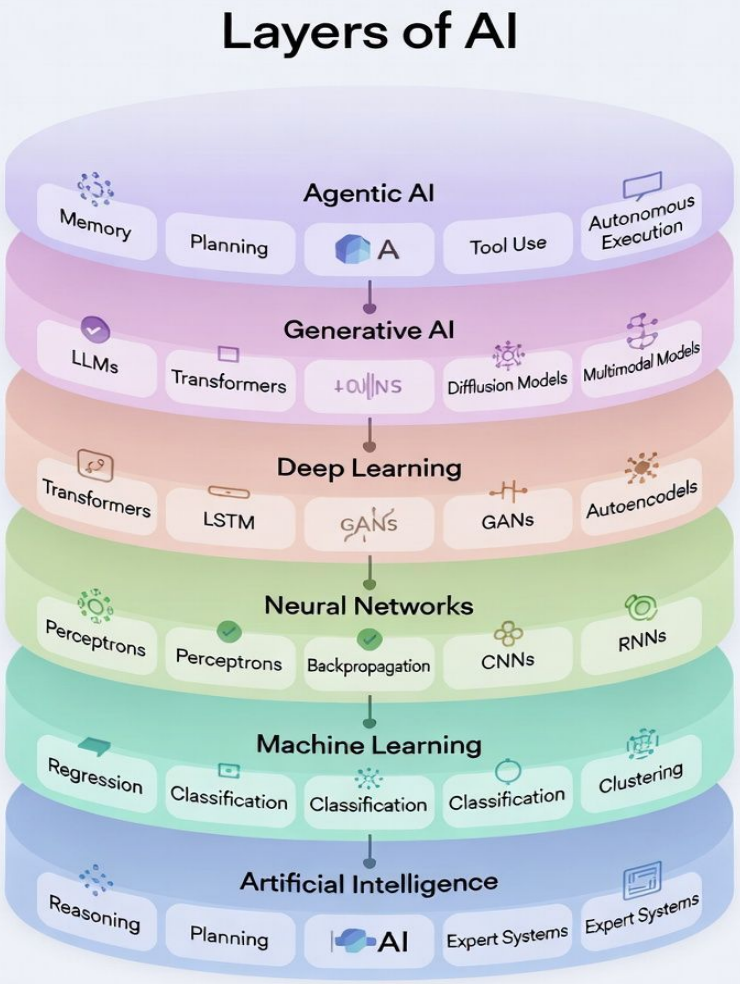
Architectures

- Fully Convolutional Networks
- SegNet
- U-Net
- RefineNet
- Learning Deconvolution Network
- Mark R-CNN



Applications: Free Space Detection





AI Evolution

- Machine learning didn’t disappear when deep learning arrived.
- Neural networks didn’t become irrelevant when transformers became popular.
- We often talk about the top layer first: agents, tools, automation, systems that appear to “think” or act on their own.
- But those ideas only make sense if you understand what sits underneath them.

Jetson AI Fundamentals

Getting Started with AI on Jetson Nano

System Setup

Camera Capture

Classification

Regression

Hello AI World

Image Recognition

Object Detection

Data Collection + Training

Semantic Segmentation

JetBot

Hardware Setup

Software Setup

Collision Avoidance

Road Following

ELE495

- Capstone Project
- 2223-Yaz: JetBot Material Classification
  - https://youtu.be/4ZHhM5hg72c
- 2324-Bahar: Conveyor Belt System, Robot Arm Project
- 2324-Yaz: JetBot Autonomous Parking
  - https://github.com/orgs/ELE495-2324Summer/repositories
- 2425-Bahar: Autonomous RF Emitter Location Finding
  - https://github.com/orgs/ELE495-2425Spring/repositories
  - Grup alfa: https://www.youtube.com/watch?v=lptr_yDA2u8
- 2425-Yaz: Voice Controlled Autonomous Vehicle
  - https://github.com/orgs/ELE495-2425Summer/repositories
- 2526-Bahar: Production with Robot Arm

Thanks!

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